

Vamshi Krishna Perabathula

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WORK EXPERIENCE

Underwriters Laboratories Inc, SAR Data Intern

05/2024 – 08/2024 | Fremont, California

- Collected and analyzed large-scale SAR datasets using statistical models and Python, ensuring compliance with KDB standards.
- Automated data workflows for test file generation, boosting operational efficiency by 20%.
- Prepared technical reports summarizing key insights from data trends, aiding stakeholder decision-making.
- Collaborated with cross-functional teams to integrate testing data and improve data-driven software-hardware synchronization.

SOLBOTS ROBOTIC SOLUTIONS, Project Intern

09/2022 – 03/2023 | Hyderabad, India

- Developed to analyze IoT sensor data, enhancing predictive maintenance and performance monitoring.
- Automated real-time data pipelines on Raspberry Pi, improving response times by 30%.
- Provided data insights through reports and visualizations for system optimization and decision-making.

TheSmartBridge, IoT Intern

06/2019 – 07/2019 | Hyderabad, India

- Designed an IoT solution using Machine Learning algorithms, Python, and IBM Cloud to analyze system data and detect anomalies, reducing errors by 20%.
- Built dynamic dashboards to visualize sensor data, supporting operational improvements in smart home automation.
- Documented project outcomes and recommended data-driven improvements to enhance system reliability.

PROJECTS

BIKE SHARING DEMAND PREDICTION

- Built machine learning models using Python and scikit-learn to forecast bike rental demand based on 2+ years of historical data.
- Identified factors like weather and seasonality, optimizing resource allocation by 30%.
- Delivered insights to optimize resource allocation and operational efficiency for bike-sharing services.

CUSTOMER CHURN PREDICTION IN BANKING

- Developed classification models with 85% accuracy using Python and scikit-learn to predict customer churn.
- Conducted EDA and feature engineering, reducing churn rates by 10% in test simulations.
- Generated dashboards to communicate insights for improving customer satisfaction metrics.

NLP SENTIMENT ANALYSIS AND CLASSIFICATION

- Processed 10,000+ text samples for sentiment analysis using TF-IDF, word embeddings, and classification models.
- Improved model accuracy by 20% using optimized deep learning models (TensorFlow).
- Delivered insights through sentiment trends and visualizations for targeted marketing decisions.

VIDEO GAME SALES ANALYSIS

- Analyzed 3,000+ records to identify trends across regions and genres using SQL and Python.
- Created predictive models to improve sales forecasting accuracy by 15%.
- Developed visual dashboards to assist publishers in optimizing their marketing strategies.

SMART SECURITY SYSTEMS FOR HOME

- Engineered a scalable IoT-based home security system with real-time monitoring using Raspberry Pi, Python, and cloud processing.
- Integrated facial recognition with 95% accuracy using OpenCV, improving home security measures.
- Developed secure keyless entry mechanisms, ensuring user safety and data protection.

SKILLS

- Technical Skills : Python, R, SQL, C
- Concepts: NLP, Machine Learning, Statistical Modeling, Predictive analysis
- Libraries & Frameworks : TensorFlow, Keras, OpenCV, Scikit-learn, Pandas, NumPy
- Data Management & Visualization: ,MySQL,PostgreSQL, Tableau ,Power BI,Matplotlib
- Tools & Platforms: Jupyter Notebooks, Git, IBM Cloud

EDUCATION

University of the Pacific,
Masters of Science in Data Science | CGPA : 3.44

08/2023 – 05/2025 | San Francisco, CA

Jawaharlal Nehru Technological University, Bachelor Of
Technology in Electronics and Communication Engineering

06/2018 – 08/2022 | Hyderabad, India